INCH-POUND

MIL-DTL-3922/70C
27 July 2010
SUPERSEDING
MIL-DTL-3922/70B
6 March 2000

DETAIL SPECIFICATION SHEET

FLANGES, WAVEGUIDE (COVER)
(SQUARE, 4 HOLE, BUTT MOUNT)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the flanges described herein shall consist of this specification sheet and MIL-DTL-3922.

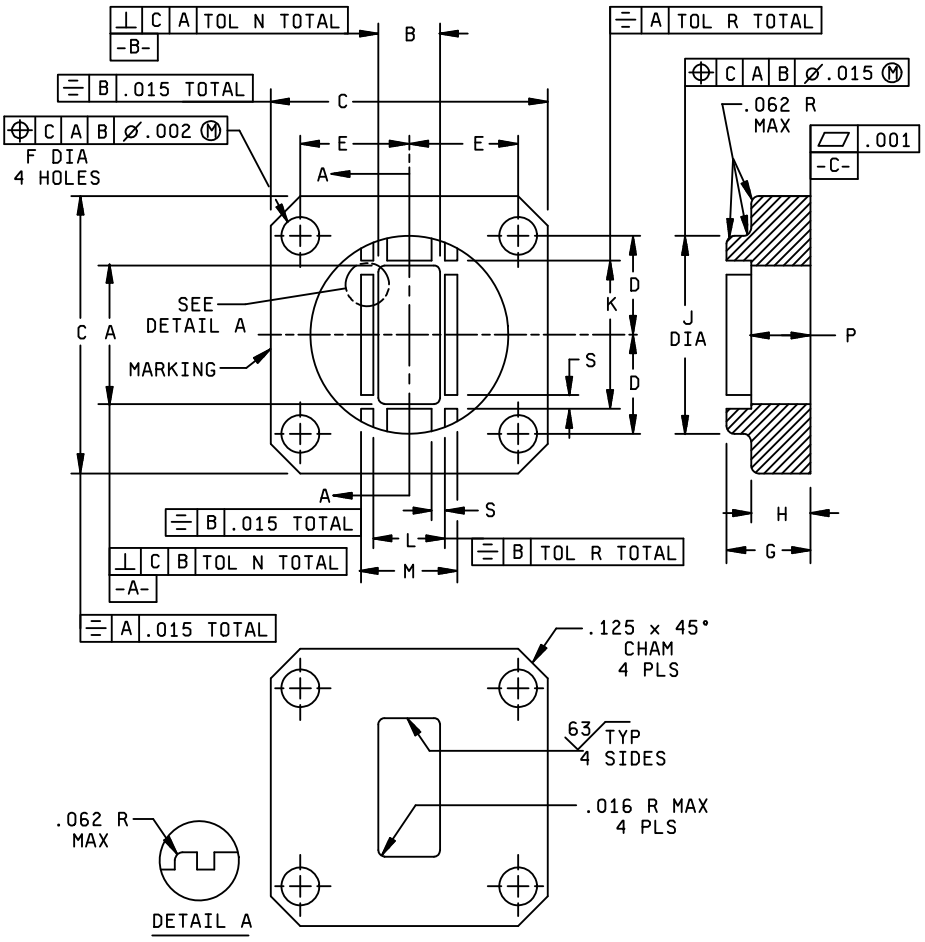


FIGURE 1. Flange configuration.

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Inches	mm
.001	0.03
.002	0.05
.015	0.38
.016	0.41
.062	1.57
.094	2.39
.125	3.18

NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Roughness of mating surfaces shall not exceed 63 microinches in accordance with ASME-B46.1,except that flaws shall be included in the roughness height measurement.
4. End item offered shall be of single-piece construction.

FIGURE 1. Flange configuration - Continued.

TABLE I. Dash numbers and dimensions.

PIN M3922/70-						Dimensions 1/ 2/ 3/	
Dash No. 4/	Used with		Dash No. 5/	Used with		A	B
	Waveguide M85/1-	Flange M3922/		Waveguide M85/1-	Flange M3922/		
001	155 156	69-001	002	157, 158, 160	69-002, 69-003	1.020 ±.003 (25.91) (0.08)	.510 ±.003 (12.95) (0.08)
004	081 085	59-010	005	083, 084, 179	59-011	.750 ±.003 (19.05) (0.08)	.375 ±.003 (9.52) (0.08)
007	087, 089, 093	59-001	008	090, 091, 180	59-002	.622 ±.002 (15.80) (0.05)	.311 ±.002 (7.90) (0.05)
010	094 076	69-004	011	097, 098, 181	69-005, 69-006	.510 ±.002 (12.95) (0.05)	.255 ±.002 (6.48) (0.05)
013	155 156	69-001	014	157, 158, 160	69-002, 69-003	1.020 ±.003 (25.91) (0.08)	.510 ±.003 (12.95) (0.08)
016	081 085	59-010	017	083, 084, 179	59-011	.750 ±.003 (19.05) (0.08)	.375 ±.003 (9.52) (0.08)
019	087, 089, 093	59-001	020	090, 091, 180	59-002	.622 ±.002 (15.80) (0.05)	.311 ±.002 (7.90) (0.05)
022	094 096	69-004	023	097, 098, 181	69-005, 69-006	.510 ±.002 (12.95) (0.05)	.255 ±.002 (6.48) (0.05)
025	081 085	59-010	026	083, 084, 179	59-011	.750 ±.003 (19.05) (0.08)	.375 ±.003 (9.52) (0.08)
027	100, 102, 106	59-003	028	103, 104, 182	59-004	.420 ±.002 (10.67) (0.05)	.170 ±.002 (4.32) (0.05)

See footnotes at end of table.

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TABLE I. Dash numbers and dimensions - Continued.

Dash No. 4/	Dash No. 5/	Dimensions 1/ 2/ 3/						
		C ± .015 (0.38)	D BSC	E BSC	F +.003, -.000 (0.08) (0.00)	G ± .015 (0.38)	H ± .015 (0.38)	J ± .015 (0.38)
001	002	1.687 (42.85)	.640 (16.26)	.670 (17.02)	.169 (4.29)	.500 (12.70)	.250 (6.35)	1.485 (37.72)
004	005	1.500 (38.10)	.520 (13.21)	.561 (14.25)	.144 (3.66)	.500 (12.70)	.188 (4.78)	1.156 (29.36)
007	008	1.312 (33.32)	.497 (12.62)	.478 (12.14)	.144 (3.66)	.375 (9.52)	.188 (4.78)	1.000 (25.40)
010	011	1.312 (33.32)	.478 (12.14)	.497 (12.62)	.144 (3.66)	.375 (9.52)	.250 (6.35)	1.000 (25.40)
013	014	1.687 (42.87)	.640 (16.26)	.670 (17.02)	.169 (4.29)	.500 (12.70)	.250 (6.35)	1.485 (37.72)
016	017	1.500 (38.10)	.520 (13.21)	.561 (14.25)	.144 (3.66)	.500 (12.70)	.250 (6.35)	1.156 (29.36)
019	020	1.312 (33.32)	.497 (12.62)	.478 (12.14)	.144 (3.66)	.375 (9.52)	.187 (4.75)	1.000 (25.40)
022	023	1.312 (33.32)	.478 (12.14)	.497 (12.62)	.144 (3.66)	.375 (9.52)	.187 (4.75)	1.000 (25.40)
025	026	1.500 (38.10)	.520 (13.21)	.561 (14.25)	.144 (3.66)	.406 (10.31)	.203 (5.16)	1.156 (29.36)
027	028	.875 (22.22)	.320 (8.13)	.335 (8.51)	.116 (2.95)	.285 (7.24)	.156 (3.96)	.625 (15.88)

See footnotes at end of table.

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TABLE I. Dash numbers and dimensions - Continued.

Dash No. 4/	Dash No. 5/	Dimensions 1/ 2/ 3/						
		K .002 +.003, -.000 (0.05) (0.08) (0.00)	L	M Min	N	P ± .005 (0.13)	R	S ±.015 (.38)
001	002	1.151 +.003, -.000 (29.24) (0.08) (0.00)	.641 +.003, -.000 (16.28) (0.08) (0.00)	.875 (22.22)	.003 (0.08)	.250 (6.35)	.002 (0.05)	.094 (2.39)
004	005	.853 +.003, -.000 (21.67) (0.08) (0.00)	.478 +.003, -.000 (12.14) (0.08) (0.00)	1.156 (29.36)	.003 (0.08)	.188 (4.78)	.002 (0.05)	.094 (2.39)
007	008	.705 +.002, -.000 (17.91) (0.05) (0.00)	.394 +.002, -.000 (10.01) (0.05) (0.00)	1.000 (25.40)	.002 (0.05)	.188 (4.78)	.001 (0.03)	.094 (2.39)
010	011	.593 +.002, -.000 (15.06) (0.05) (0.00)	.338 +.002, -.000 (8.59) (0.05) (0.00)	1.000 (25.40)	.002 (0.05)	.250 (6.35)	.001 (0.03)	.094 (2.39)
013	014	1.151 +.003, -.000 (29.24) (0.08) (0.00)	.641 +.003, -.000 (16.28) (0.08) (0.00)	.875 (22.22)	.003 (0.08)	.375 (9.52)	.002 (0.05)	.094 (2.39)
016	017	.853 +.003, -.000 (21.67) (0.08) (0.00)	.478 +.003, -.000 (12.14) (0.08) (0.00)	1.156 (29.36)	.003 (0.08)	.281 (7.14)	.002 (0.05)	.094 (2.39)
019	020	.705 +.002, -.000 (17.91) (0.05) (0.00)	.394 +.002, -.000 (10.01) (0.05) (0.00)	1.000 (25.40)	.002 (0.05)	.250 (6.35)	.001 (0.03)	.094 (2.39)
022	023	.593 +.002, -.000 (15.06) (0.05) (0.00)	.338 +.002, -.000 (8.59) (0.05) (0.00)	1.000 (25.40)	.002 (0.05)	.250 (6.35)	.001 (0.03)	.094 (2.39)
025	026	.853 +.003, -.000 (21.67) (0.08) (0.00)	.478 +.003, -.000 (12.14) (0.08) (0.00)	1.156 (29.36)	.003 (0.08)	.281 (7.14)	.002 (0.05)	.094 (2.39)
027	028	.503 +.002, -.000 (12.78) (0.05) (0.00)	.253 +.002, -.000 (6.43) (0.05) (0.00)	.625 (15.88)	.002 (0.05)	.160 (4.06)	.001 (0.03)	.094 (2.39)

1/ Dimensions are in inches.
2/ Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3/ Metric equivalents are in parentheses.
4/ Material shall be copper alloy.
5/ Material shall be aluminum alloy.

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REQUIREMENTS:

Dimensions and configuration: See figure 1 and table I.

Material: Metallic alloy as specified in table I.

Marking: See figure 1.

Part or Identifying Number (PIN): M3922/70- (dash number from table I).

Cross reference and engineering information: See table II.

TABLE II. Cross reference and engineering information.

PIN M3922/70-	AN nomenclature	Frequency range (GHz)
001	UG-1493/U	7.0 - 11.0
002, 013, 014	---	7.0 - 11.0
004, 005, 016, 017, 025, 026	---	10.0 - 15.0
007, 008, 019, 020	---	12.4 - 18.0
010, 011, 022, 023	---	15.0 - 22.0
027, 028	---	18.0 - 26.5

NOTE: Dash numbers 003, 006, 009, 012, 015, 018, 021, and 024 have been replaced by dash numbers 002, 005, 008, 011, 014, 017, 020, and 023, respectively. A separate PIN for parts constructed from aluminum alloy 6063 is no longer required.

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Referenced documents. In addition to MIL-DTL-3922 this specification sheet references the following documents:
ASME-B46.1

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